# Summary of paper 01: tables and figures toward an optimal data-generating process

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November 09, 2026

## Problem and approach

Validating a predictive biomarker requires an interaction test. With two equal subgroups and an interaction as large as the main effect, it needs about four times the sample of a main-effect test. Aggregated N-of-1 designs have been proposed to make it affordable, and their power must be estimated by simulation. Paper 01 asks what the data-generating process (DGP) of such a simulation should be. It starts from the DGP of Hendrickson et al. (2020), the starting DGP, in which the predictive biomarker is a correlation with the drug response. It takes the elements of the DGP one at a time, compares the candidate choices, and judges them by five criteria: existence, identifiability under carryover, fidelity to the pharmacology, analyzability and calibration status. The choices are collected as the recommended DGP, a step toward an optimal one. The Hybrid design is the primary case. All simulations use the starting response and baseline parameters of Hendrickson et al. This summary reproduces every table and figure of the paper in the paper's order. Each is generated by the same code as in the paper.

## Methods

Table 1 states which analysis targets which quantity. The paired-difference statistic E9 is the one with a closed form.

Table 1: The analyses used in this paper and the quantity each estimates.

| Analysis | Drug indicator | Estimates | Used in |
|---|---|---|---|
| Paired-difference statistic E9, E9b (closed form) | binary | on-minus-off contrast, $\beta_{bm:D}$ times the on/off correlation difference | Stages 1, 2, 4, 5; planning |
| The binary-indicator analysis (mixed model) | binary | the same contrast | Stage 2 (starting code) |
| Stress test: the binary-indicator analysis | binary | the on-minus-off contrast | Stage 6 |
| Stress test: fixed-half-life analyses | $D_{bc,it}$ at an analysis half-life of 0.5 week, the true half-life except in the cells that vary it | on-drug moderation $\beta_{bm:D}$ where the half-lives match | Stage 6 |
| Stress test: analyses with the half-life chosen by AIC | $D_{bc,it}$ at the AIC-chosen half-life (0, 0.25, 0.5, 1 or 2 weeks) | on-drug moderation $\beta_{bm:D}$ where the chosen half-life is the true one | Stage 6 |

Table 2 lays out the three simulation studies. They verify the closed forms, rerun the starting code and stress-test ten analyses under a prognostic biomarker. A fourth run simulates the analysis G10 for the planning section.

Table 2: Design of the three simulation studies in the ADEMP structure. MCSE: Monte Carlo standard error of a rejection rate, $\sqrt{p(1-p)/R}$ for $R$ replicates.

| | 1. Closed-form verification | 2. The starting code | 3. Stress test under a prognostic biomarker |
|---|---|---|---|
| **Aim** | Check the closed-form moments and power of Appendix A, and the loss of power under the constant rule | Show on the starting code that the correlation rule alone determines the loss of power | Find which analyses keep their nominal size under features the recommended DGP adds |

| | 1. Closed-form verification | 2. The starting code | 3. Stress test under a prognostic biomarker |
|---|---|---|---|
| **Data-generating mechanisms** | The correlation encoding with the constant rule and with the decaying correlation; compound symmetry and the separable AR(1) form; carryover on the drug-response mean decaying from the last on-drug value; Hybrid, crossover and OL+BDC designs; $N = 70$; grids of $c_{bm}$ and $t_{1/2}$ | The original `generateData()` at commit 58b32a9, as published and with the decaying correlation only (Appendix B.1); $c_{bm} = 0.25$. (a) Hybrid, $N = 70$, no censoring, $t_{1/2} \in \{0, 0.1, 0.2, 1\}$; (b) four designs, $N \in \{35, 70\}$, $t_{1/2} \in \{0, 0.1, 0.2\}$, the starting censoring patterns | The recommended DGP (Table 12), run three times: the reported run as defined, the first with the compounding carryover rule and the second with randomized allocation; base cell separable AR(1), $\rho = 0.7$, $N = 70$, $t_{1/2} = 0.5$ week; 15 null cells (11 without and 4 with a prognostic pathway of strength 0.3) and 26 alternative cells: power curves under the separable form ($c_{bm}$ from $-0.30$ to 0.45, 7 cells) and under compound symmetry ($\rho_{cs} = 0.8$, $c_{bm}$ from $-0.20$ to 0.25, 6 cells), and 13 cells at $c_{bm} = 0.30$ that each vary one feature ($N$, half-life, decay form, dropout, measurement error or a prognostic pathway) |
| **Estimand** | $\beta_{bm:D}$, and the mean of the paired-difference statistic | $\beta_{bm:D}$ | $\beta_{bm:D}$ |

| | 1. Closed-form verification | 2. The starting code | 3. Stress test under a prognostic biomarker |
|---|---|---|---|
| **Methods** | Paired-difference statistics E9 and E9b, closed form and simulated; the binary-indicator mixed model fitted with `lmer` | The binary-indicator analysis pipeline (`lmer`, model-based test) on both versions | Fourteen analyses fitted to the same simulated trials, of which the ten in Table 9 are reported; two-sided tests at $\alpha = 0.05$ |
| **Performance measures** | Mean, SD and power of the statistic; agreement as (simulated − closed form)/MCSE | Power; mean interaction estimate; number of repaired correlation matrices | Null rejection rate; power; failed fits |
| **Replicates** | 5,000 per cell (MCSE of power at most 0.007); 500 to 1,000 for `lmer` | (a) 500 per cell (MCSE at most 0.022); (b) 250 per cell (at most 0.032) | 1,000 per null cell (MCSE 0.0069 at a rate of 0.05), 500 per alternative cell (at most 0.022) |

# Results

### Stage 1. Feasibility of the joint distribution

Table 3 counts correlation matrices that are not positive definite. The starting code repaired 20 matrices (25%) on the grid of the earlier study without a warning. The recommended DGP stops instead, and it finds 75% of the extended grid infeasible at $c_{bm} = 0.6$. The choice rests on the existence criterion.

Table 3: Correlation matrices that are not positive definite. Figure 4 grid: open label, open label plus blinded discontinuation, crossover and Hybrid designs; $t_{1/2} \in \{0, 0.1, 0.2\}$ week. Extended grid: the last three designs; $t_{1/2} \in \{0, 0.1, 0.2, 0.5, 1\}$ week. The starting-DGP counts reproduce those of the starting code exactly. The recommended-DGP counts are the same at $\rho = 0.7$ (the recommended value) and $\rho = 0.8$ because both feasibility bounds lie between 0.3 and 0.6.

| Grid (number of matrices) | Starting DGP (constant correlation, compound symmetry, $\rho_{cs} = 0.8$) | Recommended DGP (decaying correlation, separable AR(1)), $\rho = 0.8$ | Recommended DGP, $\rho = 0.7$ |
|---|---|---|---|
| Figure 4 grid of the earlier study, $c_{bm} \in \{0, 0.3, 0.6\}$ (81) | 20 (25%) | 21 (26%) | 21 (26%) |
| Extended grid, $c_{bm} = 0.25$ (40) | 0 (0%) | 0 (0%) | 0 (0%) |
| Extended grid, $c_{bm} = 0.3$ (40) | 12 (30%) | 0 (0%) | 0 (0%) |
| Extended grid, $c_{bm} = 0.6$ (40) | 12 (30%) | 30 (75%) | 30 (75%) |

### Stage 2. The biomarker-response correlation under carryover

Table 4 gives closed-form power by half-life. Under the constant rule E9 power falls nearly to the test size at every positive half-life, and E9b keeps a little power from the baseline visit. Under the decaying rule power stays near its no-carryover value at 0.1 to 0.2 week and declines smoothly after. The choice of the decaying rule rests on identifiability under carryover.

Table 4: Closed-form power by carryover half-life (weeks). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry. Constant: the constant rule (constant correlation after first exposure). Decaying: the decaying rule (correlation decaying at the carryover half-life). E9: paired-difference statistic on post-baseline visits; E9b: the same with baseline counted as an off-drug visit, as in the binary-indicator analysis. Against Monte Carlo estimates (5,000 replicates per cell), 2 of the 24 cell-variants in this table differ by more than 2 Monte Carlo SE, and 16 of all 566 cell-variants of study 1 (about 26 expected by chance).

| Correlation rule | Statistic | 0 | 0.1 | 0.2 | 0.5 | 1 | 2 |
|---|---|---|---|---|---|---|---|
| Constant | E9 | 0.721 | 0.050 | 0.050 | 0.050 | 0.050 | 0.050 |
| | E9b | 0.654 | 0.081 | 0.081 | 0.081 | 0.081 | 0.081 |
| Decaying | E9 | 0.721 | 0.721 | 0.712 | 0.631 | 0.473 | 0.278 |
| | E9b | 0.654 | 0.653 | 0.647 | 0.586 | 0.472 | 0.327 |

Figure 1 shows the same result as lines, with Monte Carlo points on them.

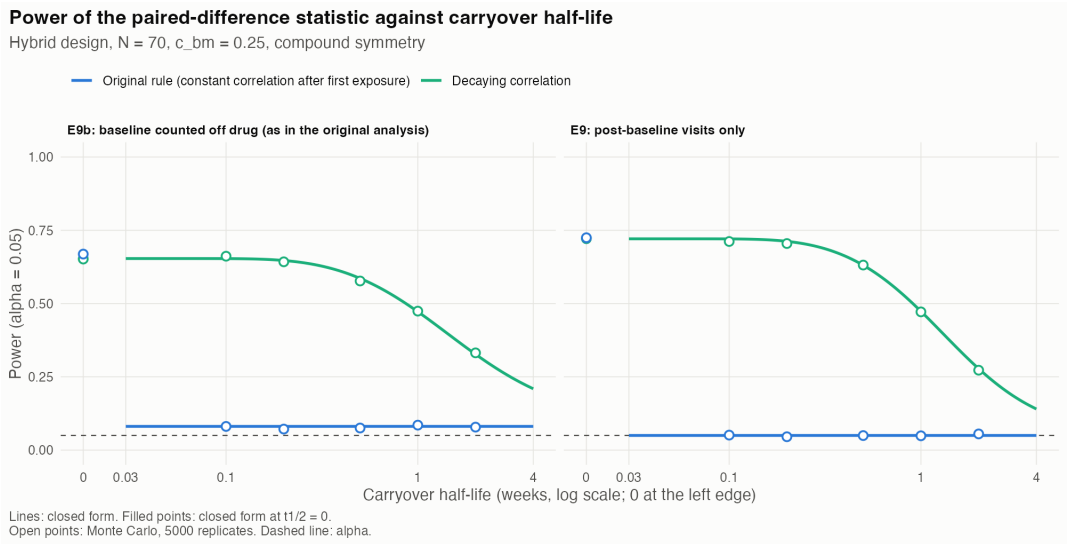



Figure 1: Closed-form power (lines) against carryover half-life, with Monte Carlo estimates (points; 5,000 replicates per cell), under the constant rule (constant correlation after first exposure) and the decaying rule (correlation decaying at the carryover half-life). Hybrid design, $N = 70$, $c_{bm} = 0.25$, compound symmetry; left panel E9b, right panel E9.

Figure 2 plots power against the on/off correlation difference. Power depends on the correlation rule only through this difference.

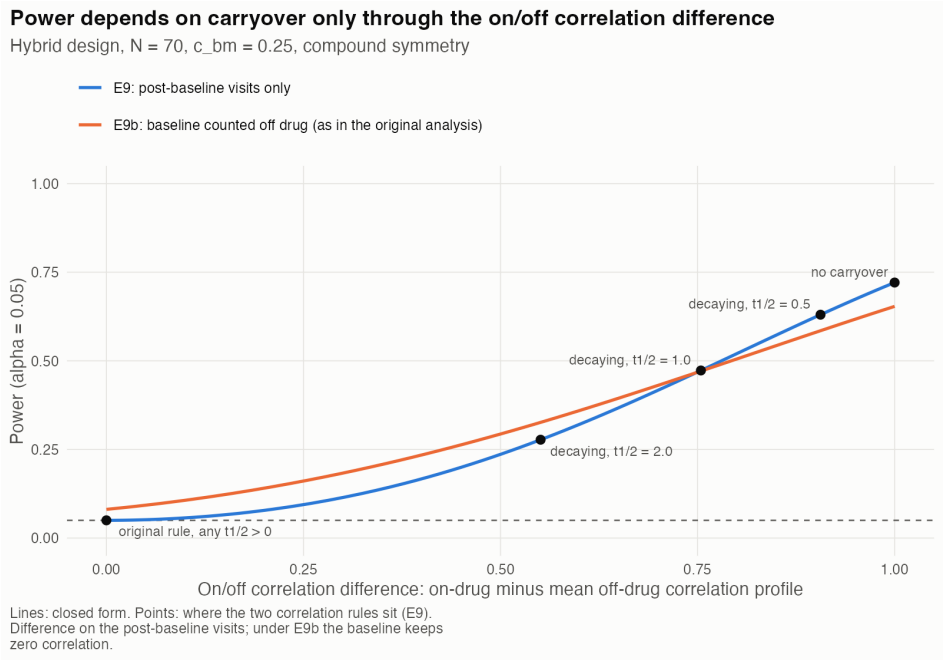



Figure 2: Closed-form power against the on/off correlation difference, with the positions of the two correlation rules. Hybrid design, $N = 70$, $c_{bm} = 0.25$, E9.

Table 5 repeats the comparison in the starting code, with only the correlation lines replaced. A rerun of the starting code under the constant rule at 0.1 week with 4,000 new replicates gave power 0.090.

Table 5: The starting code at commit 58b32a9, as published (constant) and with the decaying correlation (decaying): Hybrid design, $N = 70$, $c_{bm} = 0.25$, 500 replicates per cell, same seeds. Monte Carlo SE of power at most 0.022. A rerun of the original version at 0.1 week with 4,000 new replicates gave power 0.090 (Stage 2 text).

| $t_{1/2}$ (weeks) | Mean estimate, constant | Mean estimate, decaying | Power, constant | Power, decaying |
|---|---|---|---|---|
| 0 | -0.131 | -0.131 | 0.668 | 0.668 |
| 0.1 | -0.028 | -0.129 | 0.126 | 0.638 |
| 0.2 | -0.029 | -0.131 | 0.092 | 0.656 |
| 1 | -0.033 | -0.107 | 0.094 | 0.462 |

Figure 3 repeats the comparison for all four designs, both sample sizes and five censoring patterns. The decaying correlation keeps power where the constant rule loses it.

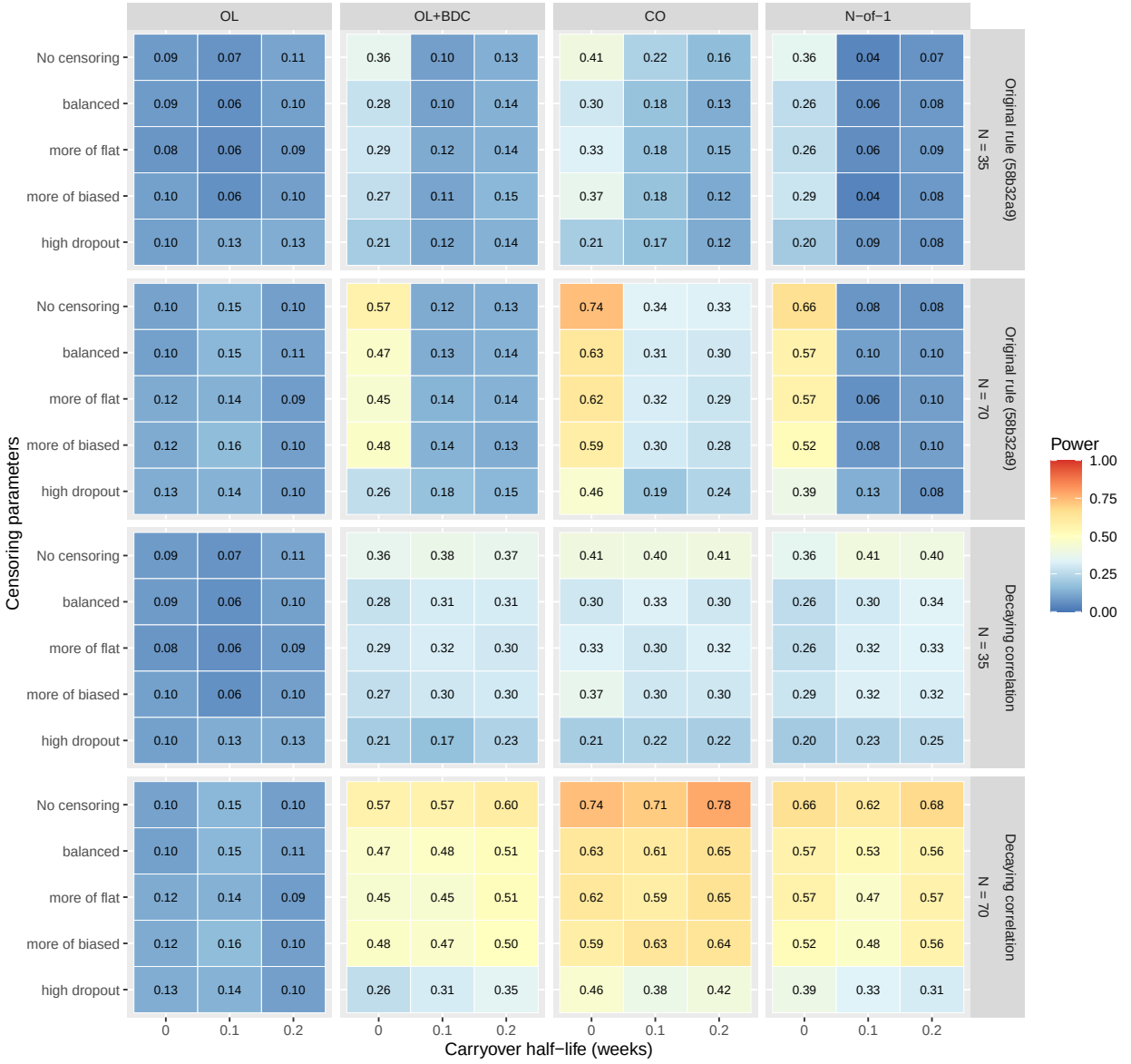



Figure 3: Power to detect the biomarker-by-treatment interaction, in the layout of Figure 4B of the earlier study, from the starting code at commit 58b32a9 as published (upper two panel rows) and with the decaying correlation (lower two panel rows). Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.032), the same seeds for both versions. No matrix was repaired in either version.

Figure 4 repeats the grid under the recommended DGP with the recommended analysis G10 (a random-intercept CAR(1) mixed model with an analysis half-life chosen by AIC on the model without the interaction, and CR2 standard errors). At $N = 70$ without censoring, Hybrid power is 0.34, 0.38 and 0.34 at half-lives of 0, 0.1 and 0.2 week. The open-label design has no off-drug visit,

so the interaction is not estimable under G10.

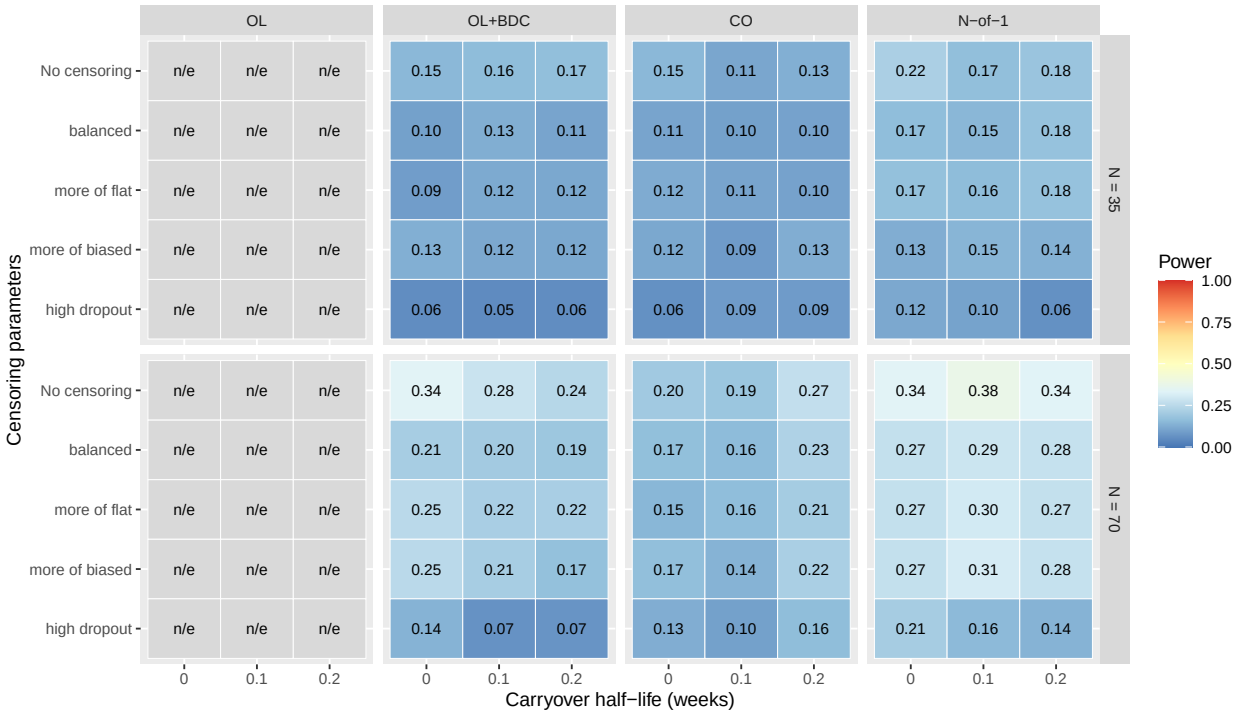



Figure 4: Power of the analysis G10 to detect the biomarker-by-treatment interaction under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, starting response parameters), in the layout of Figure 4B of the earlier study. Rows within panels: censoring patterns; columns: carryover half-life; panels: trial design (N-of-1 is the Hybrid design) and total sample size. $c_{bm} = 0.25$, 250 replicates per cell (Monte Carlo SE at most 0.031). Censoring patterns use the corrected dropout rule, under which dropout rises with worsening; the starting-code figure uses the published rule of commit 58b32a9. Gray cells (n/e): the open-label design, in which every visit is on drug and the interaction is not estimable under G10.

Figure 5 extends the closed-form comparison over interaction strengths and longer half-lives. Gray cells mark effect sizes with no valid joint distribution.

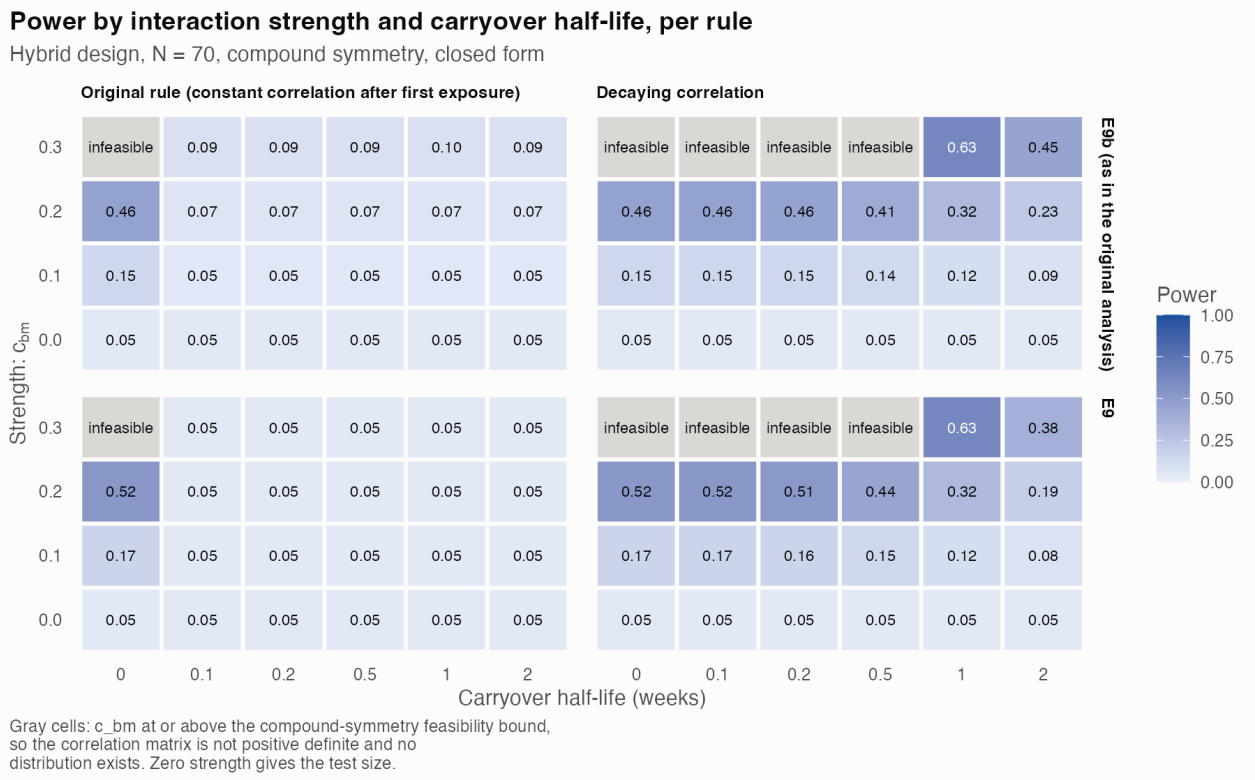



Figure 5: Closed-form power of the paired-difference statistic by interaction strength ($c_{bm}$) and carryover half-life, in the layout of Figure 4B of the earlier study: the constant rule and the decaying correlation. Upper row: E9b, which counts baseline as an off-drug visit as the binary-indicator analysis does; lower row: E9. Gray cells: $c_{bm}$ at or above the compound-symmetry feasibility bound, where the correlation matrix is not positive definite. Hybrid design, $N = 70$, compound symmetry.

## Stage 3. Carryover on the mean

Figure 6 shows the drug-response mean by visit for the four Hybrid paths under both carryover rules.

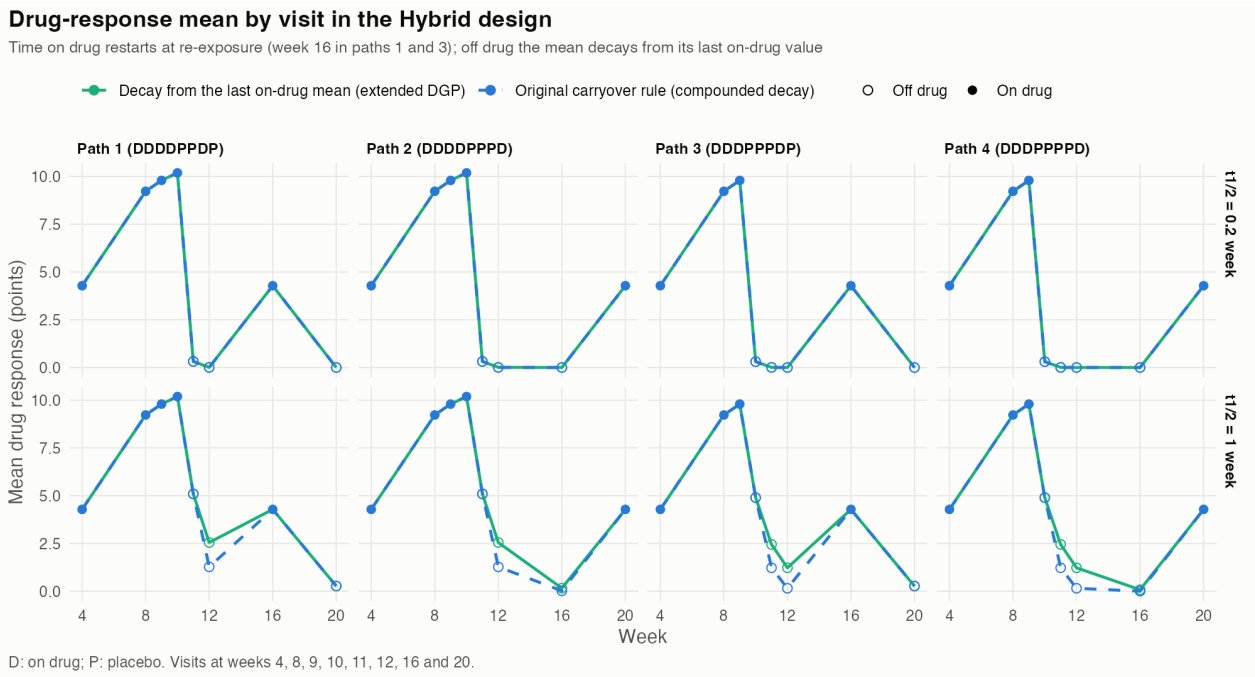



Figure 6: Drug-response mean by visit in the Hybrid design, by path, under the anchored carryover rule (decay from the last on-drug mean, the recommended DGP) and the compounding carryover rule (compounded decay), at carryover half-lives of 0.2 and 1 week. Filled points: on-drug visits; open points: off-drug visits. D: drug; P: placebo.

## Stage 4. Serial correlation

Table 6 shows what the serial correlation does to the feasibility bound and to power.

Table 6: Serial correlation: the feasibility bound on $c_{bm}$ (Ceiling; decaying correlation, no carryover, minimum over the Hybrid, crossover and OL+BDC designs) and closed-form E9 power (Hybrid, $N = 70$, no carryover) by $\rho$, under compound symmetry and under the separable AR(1) form of Stage 5. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the separable form the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho \leq 0.7$, and the crossover design is never binding. At $t_{1/2} = 0.2$ week every power value is lower by 0.004 to 0.009.

| | Compound symmetry | | Separable AR(1) | |
|---|---|---|---|---|
| $\rho_{cs}$ or $\rho$ | Ceiling | Power at $c_{bm} = 0.25$ | Ceiling | Power at $c_{bm} = 0.25$ |
| 0.5 | 0.458 | 0.492 | 0.448 | 0.261 |
| 0.6 | 0.410 | 0.553 | 0.462 | 0.270 |
| 0.7 | 0.346 | 0.628 | 0.474 | 0.290 |
| 0.8 | 0.256 | 0.721 | 0.440 | 0.335 |

Table 7 extends the separable form to $\rho$ from 0.4 to 0.95. The bound is largest at 0.7 and falls steeply above it.

Table 7: Serial correlation under the separable AR(1) form of the recommended DGP ($c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean): the feasibility bound on $c_{bm}$, minimum over the Hybrid, crossover and OL+BDC designs, by carryover half-life, the design that attains the minimum without carryover, and the closed-form total $N$ for 80 percent power of E9 in the Hybrid design at the 0.2-week half-life. Not feasible: $c_{bm}$ lies above the Hybrid bound.

| | Feasibility bound | | | | Hybrid N, E9 | |
|---|---|---|---|---|---|---|
| $\rho$ | $t_{1/2} = 0$ | 0.2 week | 1 week | Binding design | $c_{bm} = 0.25$ | $c_{bm} = 0.35$ |
| 0.40 | 0.437 | 0.438 | 0.445 | OL+BDC | 318 | 160 |
| 0.50 | 0.448 | 0.450 | 0.465 | OL+BDC | 314 | 158 |
| 0.60 | 0.462 | 0.465 | 0.494 | OL+BDC | 302 | 152 |
| 0.65 | 0.469 | 0.473 | 0.512 | OL+BDC | 292 | 148 |
| 0.70 | 0.474 | 0.479 | 0.533 | OL+BDC | 278 | 140 |
| 0.75 | 0.465 | 0.471 | 0.550 | Hybrid | 258 | 130 |
| 0.80 | 0.440 | 0.446 | 0.544 | Hybrid | 234 | 118 |
| 0.85 | 0.400 | 0.407 | 0.519 | Hybrid | 200 | 100 |
| 0.90 | 0.340 | 0.346 | 0.457 | Hybrid | 154 | not feasible |
| 0.95 | 0.247 | 0.251 | 0.343 | Hybrid | 92 | not feasible |

## Stage 5. Cross-component structure

Table 8 compares three correlation structures.

Table 8: Largest feasible biomarker-response correlation $c_{bm}$ under the decaying correlation, without carryover, minimum over the Hybrid, crossover and OL+BDC designs. Under compound symmetry the Hybrid and crossover designs tie for the minimum. Under the two AR(1) forms the minimum is attained in the Hybrid design at $\rho = 0.8$ and in the OL+BDC design at $\rho = 0.7$ (separable form: Hybrid 0.480), and the crossover design is never binding.

| $\rho_{cs}$ or $\rho$ | Compound symmetry | AR(1), two-constant cross-component | Separable AR(1) |
|---|---|---|---|
| 0.8 | 0.256 | 0.291 | 0.440 |
| 0.7 | 0.346 | 0.452 | 0.474 |

### Stage 6. Features the recommended DGP adds

Table 9 gives null rejection rates under a prognostic biomarker in the third run of study 3, the recommended DGP with fixed allocation. The first run, with the compounding carryover rule, is in Table 17.

Table 9: Null rejection rates under a prognostic biomarker ($c_{bm} = 0$). Each pathway has strength 0.3: a correlation of 0.3 with the natural-history component (constant), the placebo-response component or baseline severity, and for the growing natural-history pathway a mean shift of the natural-history component of 0.3 of its SD (3 points) per SD of biomarker at the end of its curve, growing along the curve. 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Third run of study 3: the recommended DGP, with carryover decaying from the last on-drug mean and fixed path allocation. Rates above 0.075 in bold. The first run is in Table 17; the changes between runs are given in the text.

| Analysis | Natural history, constant | Natural history, growing | Placebo response | Baseline severity |
|---|---|---|---|---|
| Binary-indicator analysis (random intercept, model-based) | **0.108** | 0.056 | **0.157** | 0.064 |
| Random intercept + CAR(1), CR2 | **0.103** | 0.049 | **0.105** | 0.052 |
| The same, half-life chosen by AIC | **0.130** | 0.059 | **0.125** | 0.057 |
| Unstructured covariance, Kenward-Roger | **0.413** | **0.200** | **0.224** | 0.048 |
| The same, half-life chosen by AIC | **0.506** | **0.258** | **0.270** | 0.054 |
| Random intercept + CAR(1) with expectancy terms, CR2 | 0.045 | 0.040 | 0.052 | 0.055 |
| The same, with phase-specific residual variances, CR2 | 0.056 | 0.037 | 0.051 | 0.059 |
| The same, with phase-specific residual variances, model-based | 0.054 | 0.042 | 0.049 | 0.059 |
| Expectancy terms, half-life chosen by AIC, CR2 | 0.058 | 0.050 | 0.060 | 0.060 |
| Random intercept + CAR(1) with free baseline slope, CR2 | 0.051 | 0.066 | 0.060 | 0.054 |

Table 10 gives the feasibility bounds when a prognostic correlation shares the biomarker row with $c_{bm}$. Each correlation lowers the room left for the other.

Table 10: Largest feasible prognostic correlation given the biomarker-response correlation, and the reverse, under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$), Hybrid design, minimum over its four paths. Correlation: the quantity whose bound is given. $c_{PB}$, $c_{TV}$, $c_{BL}$: correlation of the biomarker with the placebo-response component, the natural-history component and baseline severity. None: no value is feasible.

| Correlation | Given | $t_{1/2} = 0.2$ week | $t_{1/2} = 0.5$ week |
|---|---|---|---|
| $c_{PB}$ or $c_{TV}$ | alone | 0.513 | 0.513 |
| | $c_{bm} = 0.25$ | 0.466 | 0.477 |
| | $c_{bm} = 0.30$ | 0.435 | 0.451 |
| | $c_{bm} = 0.45$ | 0.243 | 0.306 |
| $c_{BL}$ | alone | 1.000 | 1.000 |
| | $c_{bm} = 0.25$ | 0.857 | 0.876 |
| | $c_{bm} = 0.30$ | 0.786 | 0.815 |
| | $c_{bm} = 0.45$ | 0.374 | 0.496 |
| $c_{bm}$ | alone | 0.485 | 0.518 |
| | $c_{PB} = 0.3$ | 0.422 | 0.454 |
| | $c_{PB} = 0.6$ | none | none |

Table 11 shows what each optional feature does to size and power.

Table 11: The optional features of the recommended DGP: null rejection rate ($c_{bm} = 0$; 1,000 replicates, MCSE 0.007) and power ($c_{bm} = 0.30$; 500 replicates, MCSE at most 0.022) for the binary-indicator analysis (Binary), the random-intercept CAR(1) model with CR2 and the expectancy-term model with CR2 (Expect.). Hybrid design, third run of study 3 (the recommended DGP: carryover decaying from the last on-drug mean, fixed allocation). Each row changes one feature of the base cell.

| | Null rejection | | | Power | | |
|---|---|---|---|---|---|---|
| Cell | Binary | CAR(1) | Expect. | Binary | CAR(1) | Expect. |
| Base cell ($N = 70$, $t_{1/2} = 0.5$, exponential, no dropout) | 0.077 | 0.051 | 0.067 | 0.474 | 0.486 | 0.404 |
| Measurement error (SD 3 and 6) | 0.069 | 0.050 | 0.037 | 0.422 | 0.394 | 0.314 |
| Dropout 20 percent, completely at random | 0.078 | 0.053 | 0.055 | 0.466 | 0.464 | 0.392 |
| Dropout 20 percent, dependent on last score | 0.069 | 0.060 | 0.047 | 0.424 | 0.426 | 0.360 |
| Weibull decay, shape 0.5 | 0.072 | 0.046 | 0.044 | 0.378 | 0.400 | 0.300 |
| Weibull decay, shape 2 | 0.066 | 0.039 | 0.046 | 0.526 | 0.540 | 0.496 |
| $t_{1/2} = 0$ | 0.071 | 0.040 | 0.041 | 0.536 | 0.566 | 0.498 |
| $t_{1/2} = 1$ week | 0.066 | 0.047 | 0.061 | 0.418 | 0.416 | 0.348 |
| $N = 35$ | 0.077 | 0.053 | 0.048 | 0.270 | 0.246 | 0.206 |
| $N = 150$ | 0.060 | 0.048 | 0.045 | 0.768 | 0.790 | 0.688 |

## Stage 7. The recommended DGP

Table 12 collects the choices. Each row is marked as a change from the starting DGP, a choice among open alternatives, an option, or a retained element, and names the stage that justifies it. Values not listed are the starting values.

Table 12: The recommended data-generating process. The Default column gives the value of each element in the recommended DGP; where the stress test of Stage 6 used another value it is stated. The stress test's reported run uses fixed allocation; its second run assigned participants to paths at random, a driver option outside the recommended DGP (Appendix B.5).

| Element | Specification | Default | Kind | Justified in |
|---|---|---|---|---|
| Components | $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$, Gompertz means as in the starting DGP | starting parameters (Methods): BR $m = 10.99$, $r = 0.42$, SD 8; PB and TV $m = 6.51$, $r = 0.35$, SD $10\eta_t$ and 10; $d = 5$; biomarker 124.3 (SD 15.4); baseline 83.1 (SD 18.5) | retained | Methods |
| Feasibility | stop if the correlation matrix is not positive definite | always | change | Stage 1 |
| Biomarker-response correlation | $c_{bm}$ on drug; $c_{bm}2^{-t_{sd}/t_{1/2}}$ off drug after exposure; 0 before exposure | $c_{bm}$ below the feasibility bound of every design in the study | change | Stage 2 |
| Carryover on the mean | decay from the last on-drug value, $\mu_{BR,\mathrm{last}}\, 2^{-t_{sd}/t_{1/2}}$ | $t_{1/2} = 0.1$ to 0.2 week (pharmacological, assumed); 0.5 in the stress test | choice | Stage 3 |

| Element | Specification | Default | Kind | Justified in |
|---|---|---|---|---|
| Re-exposure | as in the starting DGP: time on drug restarts at zero | on, for titrated drugs | retained | Stage 3 |
| Within-component correlation | AR(1) in weeks, $\rho^{\lvert w_t - w_s \rvert}$, one $\rho$ for all three components | $\rho = 0.7$ (convention; report and vary) | choice | Stage 4 |
| Cross-component correlation | separable, $c_1\rho^{\lvert w_t - w_s \rvert}$ at every lag; the Kronecker form applies to the correlation, since the PB SD scales with the expectancy | $c_1 = 0.2$ (starting value) | choice | Stage 5 |
| Path allocation | fixed numbers, as in the starting DGP | starting values | retained | Methods |
| Prognostic correlations | correlation of the biomarker with TV or PB, or with baseline severity; or a mean shift of TV growing along its curve | off; 0.3 in the stress test; correlations must lie inside the joint bound with $c_{bm}$ (Table 10) | option | Stage 6 |
| Measurement error | independent, SD by phase | off; 3 open label, 6 blinded in the stress test | option (driver only) | Stage 6 |
| Decay form | exponential or Weibull with the same half-life | exponential | option | Stage 6 |
| Dropout | monotone, completely at random or at random given the last score | off; 20% in the stress test | option | Stage 6 |

## Using the recommended DGP to plan a trial

Table 13 gives the total sample size for 80 percent power of the paired-difference statistic E9, and the simulated power of the recommended analysis G10 at $N = 70$. Table 14 gives G10 in every cell. G10 rejected the null at 0.046 to 0.053. At the E9 sample size for 80 percent power, G10 power was 0.844 to 0.936 in the 20 uncapped cells, so the closed-form E9 rule is conservative for G10 in these cells. Table 15 gives the G10 sample size for 80 percent power in the Hybrid design. It is 0.71 to 0.83 times the E9 value.

Table 13: Planning under the recommended DGP (separable AR(1), $\rho = 0.7$, $c_1 = 0.2$, decaying correlation, carryover decaying from the last on-drug mean, starting response parameters): total $N$ for 80 percent power of the paired-difference statistic E9 at $\alpha = 0.05$, by design, interaction strength and carryover half-life, with the implied slope in symptom points per SD of biomarker ($\sigma_{BR} = 8$) and the power at $N = 70$ and the 0.2-week half-life for E9 (closed form) and for the analysis G10 (simulated, 1,000 replicates per cell, Monte Carlo SE at most 0.016). Participants are allocated to paths in the original fixed proportions.

| | | | N for 80 percent power | | | Power at $N = 70$, 0.2 week | |
|---|---|---|---|---|---|---|---|
| Design | $c_{bm}$ | Slope (points per SD) | $t_{1/2} = 0$ | 0.2 week | 1 week | E9 | G10 |
| Hybrid | 0.25 | 2.0 | 272 | 278 | 480 | 0.285 | 0.357 |
| | 0.35 | 2.8 | 138 | 140 | 244 | 0.503 | 0.610 |
| | 0.45 | 3.6 | 82 | 84 | 146 | 0.728 | 0.836 |
| OL+BDC | 0.25 | 2.0 | 394 | 404 | $> 800$ | 0.210 | 0.275 |
| | 0.35 | 2.8 | 200 | 206 | 452 | 0.368 | 0.507 |
| | 0.45 | 3.6 | 120 | 124 | 272 | 0.558 | 0.722 |
| Crossover | 0.25 | 2.0 | 424 | 424 | 454 | 0.202 | 0.243 |
| | 0.35 | 2.8 | 216 | 216 | 230 | 0.353 | 0.365 |
| | 0.45 | 3.6 | 130 | 130 | 138 | 0.537 | 0.592 |

Table 14: Rejection rate of the analysis G10 under the recommended DGP of the preceding planning table, by design, interaction strength and carryover half-life, at $N = 70$ and at the total $N$ the closed-form E9 rule gives for 80 percent power (the preceding table). Rows with $c_{bm} = 0$ give the size at $N = 70$ (2,000 replicates per cell, Monte Carlo SE at most 0.005); the other rows give power (1,000 replicates per cell, Monte Carlo SE at most 0.016). †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Rejection rates are over converged fits (at most 0.8 percent of replicates failed in any cell).

| | | At $N = 70$ | | | At the E9 $N$ for 80 percent power | | |
|---|---|---|---|---|---|---|---|
| Design | $c_{bm}$ | $t_{1/2} = 0$ | 0.2 week | 1 week | $t_{1/2} = 0$ | 0.2 week | 1 week |
| Hybrid | 0.00 | 0.046 | 0.051 | 0.048 | | | |
| | 0.25 | 0.367 | 0.357 | 0.217 | 0.915 | 0.900 | 0.809† |
| | 0.35 | 0.616 | 0.610 | 0.362 | 0.908 | 0.899 | 0.885 |
| | 0.45 | 0.855 | 0.836 | 0.531 | 0.894 | 0.903 | 0.874 |
| OL+BDC | 0.00 | 0.052 | 0.051 | 0.051 | | | |
| | 0.25 | 0.295 | 0.275 | 0.133 | 0.936 | 0.915† | 0.601† |
| | 0.35 | 0.495 | 0.507 | 0.235 | 0.934 | 0.936 | 0.866† |
| | 0.45 | 0.725 | 0.722 | 0.362 | 0.930 | 0.932 | 0.891 |
| Crossover | 0.00 | 0.053 | 0.046 | 0.050 | | | |
| | 0.25 | 0.187 | 0.243 | 0.220 | 0.842† | 0.840† | 0.812† |
| | 0.35 | 0.356 | 0.365 | 0.378 | 0.861 | 0.867 | 0.862 |
| | 0.45 | 0.594 | 0.592 | 0.551 | 0.860 | 0.869 | 0.844 |

Table 15: Total $N$ for 80 percent power in the Hybrid design under the recommended DGP with the starting parameters: closed-form E9 (Table 13; at $\rho = 0.5$ and 0.9, Table 16) and the analysis G10. G10 was simulated at 6 values of $N$ per cell at $\rho = 0.7$ (500 replicates each), at the same multiples of the E9 value at 0.5, and at lower multiples at 0.9, where G10 needs far fewer participants than E9. The $N$ at which a probit regression of rejection on $\log N$ reaches 0.80 is rounded up.

| $\rho$ | $c_{bm}$ | $t_{1/2}$ (weeks) | E9 (closed form) | G10 (simulated) | G10 / E9 | Simulated $N$ |
|---|---|---|---|---|---|---|
| 0.7 | 0.25 | 0 | 272 | 193 | 0.71 | 95 to 340 |
| | 0.25 | 0.2 | 278 | 205 | 0.73 | 97 to 348 |
| | 0.25 | 1 | 480 | 384 | 0.80 | 168 to 600 |
| | 0.35 | 0 | 138 | 104 | 0.75 | 48 to 172 |
| | 0.35 | 0.2 | 140 | 108 | 0.77 | 49 to 175 |
| | 0.35 | 1 | 244 | 193 | 0.79 | 85 to 305 |
| | 0.45 | 0 | 82 | 61 | 0.74 | 29 to 102 |
| | 0.45 | 0.2 | 84 | 64 | 0.76 | 29 to 105 |
| | 0.45 | 1 | 146 | 121 | 0.83 | 51 to 182 |
| 0.5 | 0.25 | 0 | 308 | 271 | 0.88 | 108 to 385 |
| | 0.25 | 0.2 | 314 | 268 | 0.85 | 110 to 392 |
| | 0.25 | 1 | 542 | 434 | 0.80 | 190 to 678 |
| | 0.35 | 0 | 156 | 142 | 0.91 | 55 to 195 |
| | 0.35 | 0.2 | 158 | 140 | 0.88 | 55 to 198 |
| | 0.35 | 1 | 276 | 244 | 0.88 | 97 to 345 |
| | 0.45 | 0 | 94 | 83 | 0.88 | 33 to 118 |
| | 0.45 | 0.2 | 96 | 86 | 0.89 | 34 to 120 |
| | 0.45 | 1 | 166 | 146 | 0.88 | 58 to 208 |
| 0.9 | 0.20 | 0 | 238 | 110 | 0.46 | 36 to 238 |
| | 0.20 | 0.2 | 242 | 117 | 0.48 | 36 to 242 |
| | 0.20 | 1 | 420 | 240 | 0.57 | 63 to 420 |
| | 0.25 | 0 | 152 | 68 | 0.45 | 23 to 152 |
| | 0.25 | 0.2 | 154 | 74 | 0.48 | 23 to 154 |
| | 0.25 | 1 | 268 | 153 | 0.57 | 40 to 268 |
| | 0.30 | 0 | 104 | 50 | 0.47 | 20 to 104 |
| | 0.30 | 0.2 | 106 | 53 | 0.50 | 20 to 106 |
| | 0.30 | 1 | 186 | 107 | 0.57 | 28 to 186 |

Table 16 repeats the planning cells at the 0.2-week half-life for $\rho = 0.5$ and 0.9. At 0.9 the Hybrid design needs about half the sample, and 8 to 12 percent of G10 fits failed in each crossover cell at $N = 70$, because the fitted serial correlation reached its limit of 1 and could not be told apart from the random intercept.

Table 16: Planning by serial correlation at the 0.2-week half-life under the recommended DGP (separable AR(1), $c_1 = 0.2$, starting response parameters): the closed-form total $N$ for 80 percent power of E9, and G10 power at $N = 70$ and at that $N$, with the percent of G10 fits that failed at $N = 70$. 1,000 replicates per cell. The rows at $\rho = 0.7$ repeat Table 13. At $\rho = 0.9$ the grid of $c_{bm}$ lies inside the lower bound. Not feasible: $c_{bm}$ lies above the bound of that design. †: the E9 sample size exceeds 400 and the cell ran at $N = 400$. Power is over the fits that succeeded.

| | | | | G10 power | | |
|---|---|---|---|---|---|---|
| $\rho$ | Design | $c_{bm}$ | E9 $N$ | At $N = 70$ | At the E9 $N$ | Failed (%) |
| 0.5 | Hybrid | 0.25 | 314 | 0.258 | 0.849 | 0 |
| | | 0.35 | 158 | 0.487 | 0.829 | 0 |
| | | 0.45 | 96 | 0.693 | 0.822 | 0 |
| | OL+BDC | 0.25 | 412 | 0.225 | 0.844† | 0 |
| | | 0.35 | 208 | 0.391 | 0.859 | 0 |
| | | 0.45 | 126 | 0.606 | 0.856 | 0 |
| | Crossover | 0.25 | 348 | 0.267 | 0.864 | 0 |
| | | 0.35 | 176 | 0.405 | 0.825 | 0 |
| | | 0.45 | 106 | 0.623 | 0.815 | 0 |
| 0.7 | Hybrid | 0.25 | 278 | 0.357 | 0.900 | 0 |
| | | 0.35 | 140 | 0.610 | 0.899 | 0 |
| | | 0.45 | 84 | 0.836 | 0.903 | 0 |
| | OL+BDC | 0.25 | 404 | 0.275 | 0.915† | 0 |
| | | 0.35 | 206 | 0.507 | 0.936 | 0 |
| | | 0.45 | 124 | 0.722 | 0.932 | 0 |
| | Crossover | 0.25 | 424 | 0.243 | 0.840† | 0 |
| | | 0.35 | 216 | 0.365 | 0.867 | 0 |
| | | 0.45 | 130 | 0.592 | 0.869 | 0 |
| 0.9 | Hybrid | 0.20 | 242 | 0.546 | 0.979 | 0 |
| | | 0.25 | 154 | 0.751 | 0.976 | 0 |
| | | 0.30 | 106 | 0.897 | 0.979 | 0 |
| | OL+BDC | 0.20 | 462 | 0.416 | 0.992† | 0 |
| | | 0.25 | 296 | 0.627 | 0.999 | 0 |
| | | 0.30 | 204 | 0.757 | 0.997 | 0 |
| | Crossover | 0.20 | 628 | 0.212 | 0.803† | 10 |
| | | 0.25 | 402 | 0.315 | 0.944† | 8 |
| | | 0.30 | 278 | 0.451 | 0.938 | 9 |

# Appendix A. Closed-form derivations

## A.5 The separable structure

Each step between visits contributes the following term to the schedule term of the feasibility bound.

| Step | Term |
|---|---|
| on to on | $(1-\rho_t)/(1+\rho_t)$ |
| on to off | $\rho_t^2/(1-\rho_t^2)$ |
| off to on | $1/(1-\rho_t^2)$ |
| off to off | $0$ |

## Appendix C. First run of the stress test

Table 17: Null rejection rates under a prognostic biomarker in the first run of study 3 (compounding carryover rule, fixed allocation), 1,000 replicates per cell (two-MCSE band 0.036 to 0.064 for a single cell). Rates above 0.075 in bold.

| Analysis | Natural history, constant | Natural history, growing | Placebo response | Baseline severity |
|---|---|---|---|---|
| Binary-indicator analysis (random intercept, model-based) | **0.107** | 0.053 | **0.159** | 0.075 |
| Random intercept + CAR(1), CR2 | **0.099** | 0.049 | **0.100** | 0.050 |
| The same, half-life chosen by AIC | **0.115** | 0.058 | **0.112** | 0.055 |
| Unstructured covariance, Kenward-Roger | **0.416** | **0.195** | **0.222** | 0.051 |
| The same, half-life chosen by AIC | **0.509** | **0.245** | **0.256** | 0.064 |
| Random intercept + CAR(1) with expectancy terms, CR2 | 0.047 | 0.046 | 0.040 | 0.040 |
| The same, with phase-specific residual variances, CR2 | 0.055 | 0.044 | 0.042 | 0.042 |
| The same, with phase-specific residual variances, model-based | 0.062 | 0.043 | 0.045 | 0.046 |
| Expectancy terms, half-life chosen by AIC, CR2 | 0.059 | 0.052 | 0.047 | 0.045 |
| Random intercept + CAR(1) with free baseline slope, CR2 | 0.049 | 0.063 | 0.049 | 0.050 |

## Conclusions

Three choices matter most. First, the biomarker-response correlation must decay after discontinuation. If it stays constant after first exposure, the interaction is absent among the post-baseline visits under any carryover, and power falls nearly to the test size. With a decaying correlation, power returns to its no-carryover value at pharmacological half-lives and declines smoothly at half-lives of 0.5 week and above. Second, the serial correlation structure sets the largest interaction the DGP can hold. A separable AR(1) structure at a serial correlation of 0.7 raises the bound from 0.256 under compound symmetry to 0.474. The value 0.7 is a convention, however. At 0.9 the bound falls to 0.340, and the Hybrid design needs 154 participants at $c_{bm} = 0.25$ instead of 278. Third, a

matrix outside the bound should be refused rather than repaired.

At the closed-form sample size, G10 power was 0.844 to 0.936 in the uncapped cells. We propose the recommended DGP as a reference for simulation studies of these designs. Every parameter is labeled as a starting value, a choice or uncalibrated. Calibrating the serial correlation, the drug-response SD and the symptom half-life is the next step toward an optimal DGP.